

**A PROJECT TITLE
ON**

Bankruptcy Prediction Using Decision Tree and Random Forest

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

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By

DEERAJ AAKIRI – 2520030113

JAISWAL REDDY BEERAM – 2520030135

SAHASRA GANDLA - 2520030108

Under the Esteemed guidance of
G.RAVINDRA BABU
Associate Professor, CSE

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Abstract

This project presents a machine learning system for predicting whether a company is likely to be bankrupt or not bankrupt using financial and operational attributes. The project uses a dataset containing 3,000 company records and 22 original attributes. The dataset was collected from an online source and analysed through data inspection, preprocessing, exploratory analysis, feature selection, model development, training, testing, and evaluation. The Company_ID attribute is treated as an identifier and is excluded from model training, resulting in 20 input features and one target variable named Bankrupt.

Two supervised machine learning algorithms, Decision Tree and Random Forest, are developed for binary classification. The data is divided into training and testing sets using an 80:20 stratified split. The models are evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. The experimental results show that both models can identify bankruptcy patterns, while Random Forest provides higher overall accuracy and ROC-AUC on the selected test set.

Keywords: Bankruptcy Prediction, Machine Learning, Decision Tree, Random Forest, Financial Risk, Binary Classification

1. Introduction

1.1 Purpose of the Project

Bankruptcy of a company can result in financial losses for owners, employees, investors, lenders, suppliers, and other stakeholders. Early identification of companies that may face financial difficulties can help stakeholders take preventive actions. Traditional financial analysis generally depends on manually examining financial ratios and historical information, which can be time-consuming when a large number of companies need to be analysed.

The purpose of this project is to develop a machine learning-based bankruptcy prediction system that can learn patterns from company financial and operational data. The system uses Decision Tree and Random Forest classification algorithms to predict whether a company belongs to the bankrupt or non-bankrupt class. The project also provides a web application through which users can enter company information and obtain a prediction.

1.2 Problem Statement

The problem addressed in this project is the binary classification of companies into Bankrupt and Not Bankrupt classes using financial and operational attributes. The expected output of the system is a binary prediction where 0 represents a company that is not bankrupt and 1 represents a bankrupt company. Decision Tree and Random Forest models are trained using historical company data and evaluated using appropriate classification metrics.

1.3 Existing Methods and Disadvantages

Traditional bankruptcy prediction methods often depend on financial-ratio analysis and statistical techniques. These methods can be useful for interpreting financial conditions, but they may require manual analysis and may not capture complex relationships among multiple financial variables. They can also become difficult to apply efficiently when the dataset contains many companies and attributes.

Single machine learning models such as Decision Trees can provide interpretable predictions, but a single tree may be sensitive to the training data and may not generalize well if it becomes too complex. Ensemble methods such as Random Forest address this limitation by combining multiple decision trees. Recent research also shows that class imbalance, feature selection, and preprocessing can significantly affect bankruptcy prediction performance.

1.4 Proposed System

The proposed system follows a complete machine learning pipeline consisting of dataset collection, data inspection, cleaning, exploratory analysis, feature selection, train-test splitting, model training, model evaluation, and prediction. The project uses Decision Tree and Random Forest algorithms for classification. Company_ID is excluded because it is an identifier rather than a predictive financial attribute. The final system also provides a Flask-based web application that displays dataset information, model performance, and an interactive bankruptcy prediction interface.

1.5 Research Objectives

1. To prepare and understand a suitable dataset for company bankruptcy prediction.
2. To perform data inspection, cleaning, and exploratory analysis.
3. To select relevant financial and operational attributes for prediction.
4. To develop and train Decision Tree and Random Forest classification models.
5. To evaluate the models using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix.
6. To develop a web application for interactive bankruptcy prediction.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Bankruptcy Prediction of Manufacturing Companies of India Post-BC: A Comparative Study Between Various Predictive Techniques	Samal Kaur, Adish Arora, Anil Kumar Deyal and Arup Mondal, 2024, Indian Journal of Finance, Vol. 18, Issue 3, pp. 25-42, DOI: 10.17008/IJOF.2024.V18I3.173615.	Difficulty in accurately predicting insolvency among Indian manufacturing companies and the need to compare different predictive techniques in the post-IBC environment.	To forecast insolvency and compare different predictive techniques for bankruptcy prediction among Indian manufacturing companies.	Comparative analysis of different bankruptcy prediction techniques using financial indicators, including machine-learning and statistical approaches.	The study shows that predictive techniques can provide useful early warning information for identifying companies at risk of bankruptcy.	Use larger Indian datasets, additional financial and non-financial variables, different sectors, longer time periods and newer explainable or ensemble methods.
2	Automated Machine Learning in Bankruptcy Prediction of Manufacturing Companies	Prasada Kumar, 2024, Vol. 232, pp. 1428-1436, DOI: 10.15388/procs.2024.01.141.	Manual model selection and parameter tuning can be time-consuming, and different automated approaches may produce different prediction performance.	To investigate the application of automated machine learning for bankruptcy prediction and compare it with conventional machine-learning methods.	AutoML and H2O approaches are compared with Linear Discriminant Analysis, Logistic Regression, Naive Bayes, CatBoost and XGBoost using financial company data and ten-fold cross-validation.	The reported H2O approach achieved an AUC of 80.17%, while CatBoost and XGBoost also performed strongly, showing the usefulness of AutoML for bankruptcy prediction.	Evaluate AutoML on larger and more diverse datasets, industries and countries, including explainable AutoML, and automated feature selection.
3	Intelligent Corporate Bankruptcy Prediction: A Data-Driven Machine Learning Approach	Shivika Hemadri, Bhushan Jadhav, Anil Deshpande, Uma Giridhar and Ashish Patel, 2024, Journal of Information Systems Engineering and Management, Vol. 9, Issue 4,	Bankruptcy datasets can contain many financial variables and class imbalance, which can reduce the effectiveness of classification models.	To identify important financial factors and improve bankruptcy prediction using feature selection, data balancing and machine learning classifiers.	Random Forest feature selection, Borderline-SMOTE and classifiers including Random Forest, Decision Tree, XGBoost, CatBoost and AdaBoost are used on the Indian bankruptcy dataset.	The reported results show improved classification after preprocessing and balancing, with Random Forest achieving the highest reported accuracy of 94% in the study.	Test the approach on different industries and countries and investigate additional feature-selection, balancing, ensemble and explainable AI methods.
4	Ensemble Classifiers for Bankruptcy Prediction Using SMOTE and RFECV	T. Mahesh, Vijayashanthi Lavanya and Anirudh Rathi, 2024, International Journal of Enterprise Network Management, Vol. 15, No. 1, pp. 109-132, DOI: 10.1504/IJENM.2024.137456.	Class imbalance and feature-selection effects can strongly influence the performance of ensemble classifiers.	To study how preprocessing, feature selection and machine-learning models under bankruptcy prediction.	SMOTE, RFECV, Random Forest, AdaBoost, GBOT, XGBoost, Majority are evaluated using ROC data from 2003-2023, with temporal information incorporated through exponentially weighted moving averages.	The study indicates that class-imbalance handling can strongly influence classifier performance and that preprocessing choices are important.	Investigate more robust combinations of preprocessing, feature selection, balancing techniques and ensemble models.
5	Machine Learning Techniques in Bankruptcy Prediction: A Systematic Literature Review	Apostolova Dzelic and Anna Riggen, 2024, Expert Systems with Applications, Vol. 255, Article 124761, DOI: 10.1016/j.eswa.2024.124761.	The large amount of bankruptcy prediction research makes it difficult to identify overall trends, methods, datasets and emerging techniques.	To systematically review empirical research on machine-learning techniques for bankruptcy prediction.	PRISMA-based systematic literature review covering 207 empirical studies on bankruptcy prediction.	The review shows increasing use of advanced and hybrid machine-learning methods and highlights the importance of data, feature selection and evaluation procedures.	Conduct standardized comparative studies and further investigate explainable AI, hybrid models, data balancing and cross-country validation.
6	Predicting U.S. Bank Failures and Stress Testing with Machine Learning Algorithms	Finance Research Letters, Vol. 75, Article 106802, 2025, DOI: 10.1016/j.frl.2025.106802.	There is a need to predict financial institution failures using financial and time-related information, including under stress conditions.	To forecast U.S. bank failures and evaluate machine-learning models under financial stress conditions.	Random Forest, Logistic Regression, XGBoost, SVM and Neural Networks are evaluated using ROC data from 2003-2023, with temporal information incorporated through exponentially weighted moving averages.	Random Forest achieved the highest overall accuracy in the reported competition, while model performance varied under stress conditions.	Include more temporal, macroeconomic and stress-related information and validate models in other countries and economic environments.

3. Methodology

3.1 Proposed System Flow Diagram

Insert the proposed-system flow diagram here as per your project.

Recommended flow: Dataset → Data Cleaning → Missing-Value Treatment → Encoding → Feature Engineering/Selection → Train/Test Split → Model Training → Hyperparameter Tuning → Prediction → Evaluation → Final Model.

3.2 Dataset Description

The dataset used in this project contains **3,000 company records and 22 original attributes**. The dataset was collected from an online source and contains financial and operational information about companies. The attributes include company age, number of employees, annual revenue, total assets, total liabilities, debt-to-equity ratio, liquidity ratios, profit margins, return ratios, revenue growth, asset turnover, working capital ratio, operating cash flow, and free cash flow. The dataset contains one identifier attribute, Company_ID, 20 input features, and one target variable, Bankrupt. The target variable is binary, where 0 represents Not Bankrupt and 1 represents Bankrupt. The dataset contains 2,062 non-bankrupt records and 938 bankrupt records.

Attributes

1. Company_ID
2. Company_Age_Years
3. Employees
4. Annual_Revenue_USD
5. Total_Assets_USD
6. Total_Liabilities_USD
7. Debt_to_Equity
8. Current_Ratio
9. Quick_Ratio
10. Cash_Ratio
11. Gross_Profit_Margin
12. Operating_Profit_Margin
13. Net_Profit_Margin
14. ROA
15. ROE
16. Interest_Coverage_Ratio
17. Revenue_Growth_Rate

18. Asset_Turnover
19. Working_Capital_Ratio
20. Operating_Cash_Flow_USD
21. Free_Cash_Flow_USD
22. Bankrupt

Attributes used for prediction: 22

Identifier excluded: Company_ID

3.3 Experimental Setup

The project is implemented using Python. The development and execution environment is Visual Studio Code on a Windows operating system. Python libraries including Pandas, NumPy, Matplotlib, Scikit-learn, and Flask are used. Pandas is used for dataset handling, NumPy for numerical operations, Matplotlib for visualization, Scikit-learn for machine learning model development and evaluation, and Flask for developing the web application.

3.4 Data Preprocessing

- Dataset inspection and schema validation.
- Duplicate-record analysis.
- Missing-value analysis.
- Numerical data validation.
- Removal of the Company_ID identifier from model features.
- Preparation of the Bankrupt target variable.
- Selection of 20 input attributes.
- Stratified train-test splitting.
- Prevention of data leakage by fitting the models only on the training data.

3.5 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the structure and distribution of the dataset. The analysis includes:

- Dataset shape analysis.
- Attribute and data-type analysis.
- Missing-value analysis.
- Duplicate-value analysis.
- Bankrupt versus non-bankrupt class distribution.
- Distribution of financial attributes.
- Correlation analysis between numerical variables.
- Feature-importance analysis using Random Forest.

3.6 Feature Engineering and Feature Selection (if applicable)

The dataset contains 22 original attributes. Company_ID was excluded because it is an identification field and does not represent a meaningful financial characteristic for predicting bankruptcy. The remaining 20 numerical attributes were selected as input features. The Bankrupt attribute was retained separately as the target variable.

No artificial feature was created in the initial model. The selected features represent company size, profitability, liquidity, leverage, growth, efficiency, and cash-flow characteristics.

3.7 Model Building and Training

Decision Tree

A Decision Tree classifier was trained to learn decision rules from the financial attributes. The model recursively divides the data based on feature conditions and produces a final classification of Bankrupt or Not Bankrupt.

Random Forest

A Random Forest classifier was developed as an ensemble model containing multiple decision trees. The model combines the predictions of multiple trees to produce the final classification. In this project, the Random Forest model uses 200 trees and class-weight balancing.

The dataset was divided into **80% training data and 20% testing data** using a stratified split.

3.8 Hyperparameter Tuning(if applicable)

For the initial implementation, controlled model parameters were used rather than performing an extensive grid search. The Decision Tree uses a maximum depth of 8, while the Random Forest uses 200 decision trees. Both models use balanced class weights to give additional consideration to the minority bankruptcy class.

Further hyperparameter optimization can be performed in future work using Grid Search or Randomized Search with cross-validation.

3.9 Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

These metrics provide a more complete evaluation than accuracy alone, especially because bankruptcy prediction involves two classes and correctly identifying bankrupt companies is important

4. Results and Discussion

4.1 Preprocessing Results

The original dataset contains **3,000 records and 22 attributes**. The dataset contains **no missing values and no duplicate records** in the current dataset. One attribute, Company_ID, was excluded because it is an identifier. Therefore, **20 attributes were used as input features**, while Bankrupt was used as the target variable.

The target distribution contains **2,062 Not Bankrupt records and 938 Bankrupt records**. An 80:20 stratified train-test split was used, resulting in **2,400 training records and 600 testing records**.

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Decision Tree	75.00%	59.22%	64.89%	61.93%	74.42%
Random Forest	81.00%	77.61%	55.32%	64.60%	85.57%

4.3 Visual Results and discussion

The following visualizations should be included in the report:

1. Dataset class-distribution chart.
2. Financial-feature distribution plots.
3. Correlation heatmap.
4. Decision Tree confusion matrix.
5. Random Forest confusion matrix.
6. Model performance comparison chart.
7. Random Forest feature-importance chart.

The confusion matrices show the numbers of correctly and incorrectly classified companies. The feature-importance visualization can be used to identify which financial attributes contribute most to the Random Forest model's decisions.

5. Conclusion

The project developed a machine learning-based system for predicting corporate bankruptcy using financial and operational attributes. The dataset contains 3,000 records and 22 original attributes, from which 20 attributes were used as input features after excluding the company identifier. Decision Tree and Random Forest models were trained and evaluated using an 80:20 stratified train-test split.

The experimental results show that both algorithms can perform binary bankruptcy classification on the selected dataset. In the current experiment, Random Forest achieved 81.00% accuracy and 85.57% ROC-AUC, while Decision Tree achieved 75.00% accuracy and 74.42% ROC-AUC. The developed Flask web application further provides an interactive interface for viewing the dataset, model results, and generating bankruptcy predictions.

6. Future Work

1. Use larger and more diverse company datasets from multiple industries and regions.
2. Perform extensive hyperparameter optimization and cross-validation.
3. Investigate additional ensemble and deep learning algorithms.
4. Apply advanced techniques for handling class imbalance.
5. Use explainable AI techniques such as SHAP to explain individual predictions.
6. Deploy the system as a real-time cloud-based application.
7. Validate the trained models using external datasets.
8. Continuously monitor model performance when new financial data becomes available.

7. References

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Signature of the Guide

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